Findings: Additional Evaluation of Credit Default Prediction

This analysis evaluates whether adding JV NBI credit information improves default prediction beyond bank information. Both models include young, female, and low-income indicators. The expanded model also includes JV NBI credit limit and balance.

The models were trained using 70% of the sample and evaluated on the remaining 30%. The test sample contains 523,437 borrowers, of whom 6.13% defaulted.

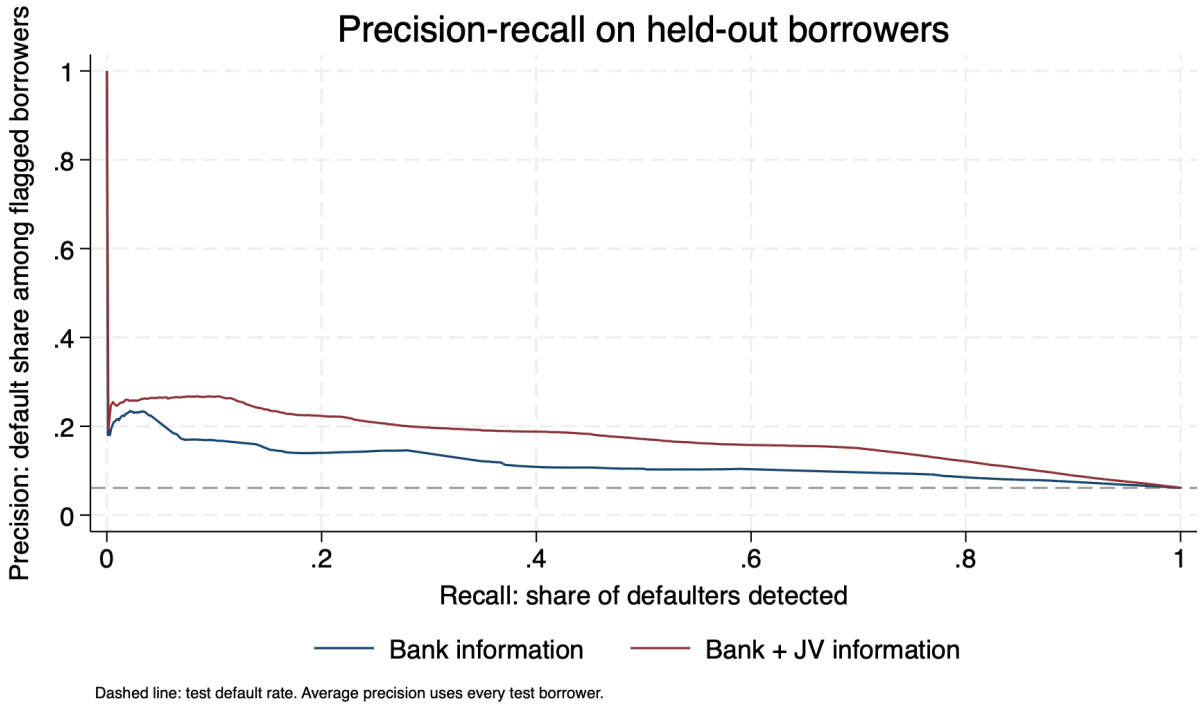
Overall Performance

Adding JV information improved all main performance measures. AUC increased from 0.674 to 0.769, indicating better separation between borrowers who defaulted and those who did not. Average precision increased from 0.115 to 0.172, an improvement of approximately 50%.

Log loss and Brier score also declined, indicating that the expanded model produced better probability predictions overall.

metric	bank	bank_jv	change	improvement
auc	0.674402631	0.7691048513	0.0947022203	0.0947022203
average_precision	0.1146058754	0.1719941018	0.05738822637	0.05738822637
brier	0.05665316679	0.05474587646	-0.001907290334	0.001907290334
lift_top05	2.628896595	3.66867704	1.039780445	1.039780445
lift_top10	2.324053775	3.183844561	0.8597907861	0.8597907861
log_loss	0.2232549424	0.2109552677	-0.01229967475	0.01229967475
n	523437	523437	0	
precision_top05	0.1611077264	0.2248290092	0.06372128281	0.06372128281
precision_top10	0.1424259213	0.1951168256	0.05269090423	0.05269090423
prevalence	0.06128340182	0.06128340182	0	
recall_fpr05	0.1422470229	0.2178128312	0.07556580834	0.07556580834
recall_fpr10	0.2600224453	0.35971694	0.09969449467	0.09969449467

The precision–recall curve shows the same pattern. At most recall levels, the expanded model identifies a group with a higher concentration of actual defaulters.



Dashed line: test default rate. Average precision uses every test borrower.

Screening Performance

Among the highest-risk 5% of borrowers, the default rate increased from 16.11% under the bank model to 22.48% under the expanded model. This means that, with the same screening capacity, JV information helps identify more borrowers who subsequently default.

A separate exercise selected screening thresholds using validation data within the training sample. At a target false-positive rate of 5%, the bank model detected 14.21% of defaulters, compared with 21.54% for the expanded model. Actual test false-positive rates were similar, at approximately 5%.

These screening models used a smaller estimation sample because part of the original training sample was reserved for selecting thresholds.

model	target_fpr	threshold	actual_test_fpr	test_recall	test_precision	flagged	defaults	n_test
bank	0.05	0.1121441964	0.04984949904	0.1421223268	0.1569201115	29053	4559	523437
bank	0.1	0.07559949731	0.0993367375	0.2582455265	0.1450940554	57094	8284	523437
jv	0.05	0.1275213509	0.04940990193	0.2154124322	0.2215595742	31188	6910	523437
jv	0.1	0.1065815639	0.09946088298	0.3577218031	0.1901534484	60346	11475	523437

Which Information Drives the Improvement?

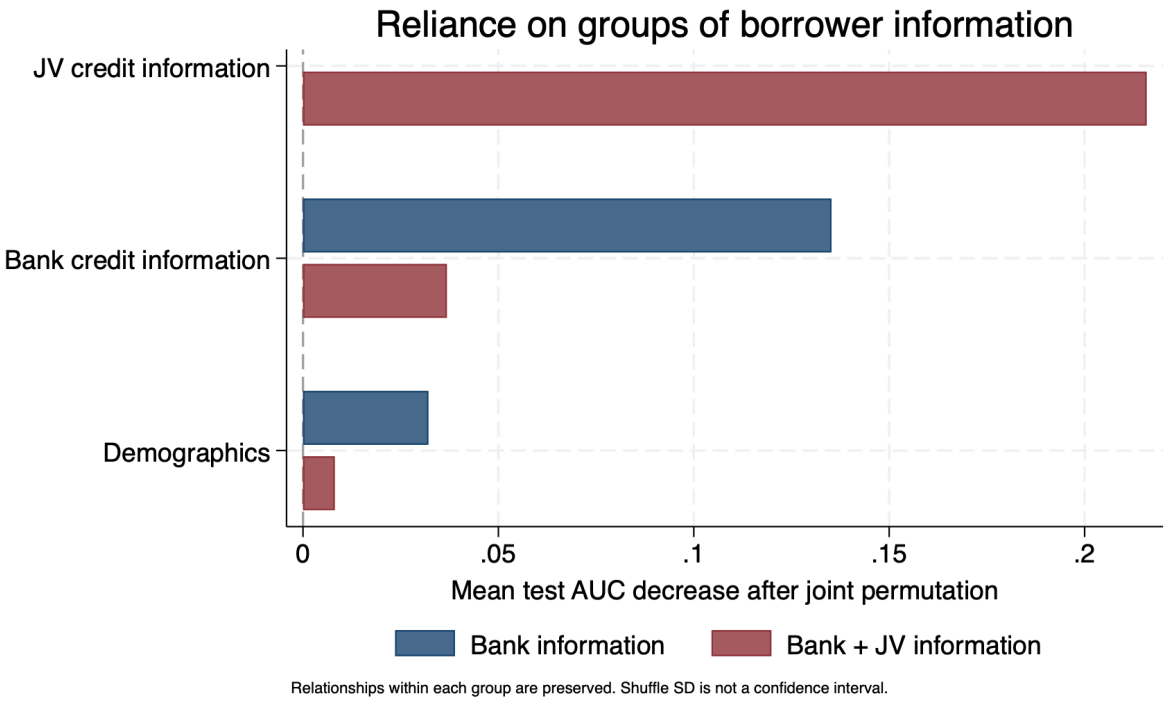
The ablation analysis compared models that added JV limit, JV balance or both to the bank model. Adding JV limit alone increased AUC to 0.689, while adding JV balance alone increased it to 0.681. Including both increased AUC to 0.769.

The much larger improvement when both variables are included suggests that credit limit and outstanding balance provide complementary information about default risk.

metric	valuebalance	valuebank	valuejv	valuelimit	gain_limit	gain_balance	gain_jv
auc	0.6807582347	0.674402631	0.7691048513	0.688549972	0.01414734107	0.00635560377	0.0947022203
average_precision	0.1171214908	0.1146058754	0.1719941018	0.1208898762	0.0062840076	0.002515615431	0.0573882637
brier	0.05655936164	0.05665316679	0.05474587646	0.05634608215	0.0003070846442	0.00009380514896	0.001907290334
lift_top05	2.625332679	2.628896595	3.66867704	2.651747697	0.02285110219	-0.003563915972	1.039780445
lift_top10	2.185639304	2.324053775	3.183844561	2.354604716	0.03055094164	-0.1384144703	0.8597907861
log_loss	0.222397668	0.2232549424	0.2109552677	0.2203958342	0.002859108223	0.0008572744274	0.0122996747 5
n	523437	523437	523437	523437			
precision_top05	0.1608893175	0.1611077264	0.2248290092	0.1625081197	0.001400393277	-0.0002184088946	0.0637212 8281
precision_top10	0.1339434117	0.1424259213	0.1951168256	0.144298187	0.001872265632	-0.0084825096	0.0526909 0423
prevalence	0.06128340182	0.06128340182	0.06128340182	0.06128340182			
recall_fpr05	0.1463308186	0.1422470229	0.2178128312	0.1445227259	0.002275702974	0.004083795748	0.0755658 0834
recall_fpr10	0.2344285803	0.2600224453	0.35971694	0.2560321716	-0.003990273708	-0.02559386495	0.0996944 9467

Grouped permutation importance supports this finding. Shuffling JV limit and balance together reduced the expanded model's AUC by approximately 0.216. Shuffling bank information reduced it by 0.037, while shuffling demographics reduced it by 0.008.

The expanded model therefore relies most heavily on JV credit information. These importance measures describe reliance on inputs; they do not establish that those characteristics cause default.

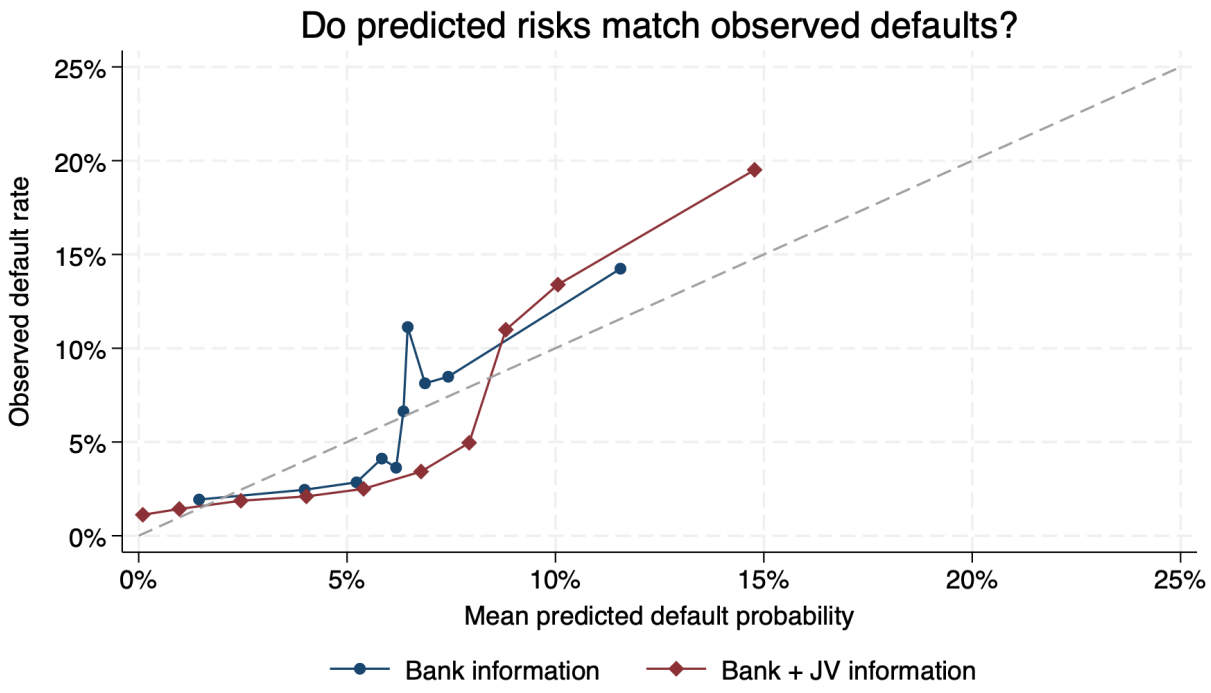


Calibration

Calibration examines whether predicted default probabilities match observed default rates. The expanded model's overall calibration intercept and slope were close to their ideal values of zero and one.

However, the calibration bins show that predicted risks still differ from observed outcomes in several parts of the sample. For the highest-risk tenth of borrowers, the expanded model predicted an average default probability of 14.78%, while the observed default rate was 19.51%. It also overpredicted default in several middle-risk groups.

The expanded model improves probability prediction overall, but further work could improve the accuracy of its risk estimates across borrower groups.



Calibration Parameters

model	intercept	intercept_se	slope	slope_se	n
bank	0.1372747202	0.0373708703	1.052181688	0.01404015036	523437
jv	0.01023542225	0.0235323916	1.004368864	0.009362547579	523437

Calibration Bins

calibration_bin	predicted	observed	n	model
1	0.0009975188247	0.011176066	52344	jv
2	0.009760751408	0.014290081	52344	jv
3	0.02452645876	0.018684089	52344	jv
4	0.04024270439	0.020996122	52343	jv
5	0.05396466017	0.025007641	52344	jv
6	0.06776922891	0.034254164	52344	jv
7	0.07929784263	0.049557727	52343	jv
8	0.08810584953	0.10985022	52344	jv
9	0.1006041673	0.13390264	52344	jv
10	0.1477878348	0.19511683	52343	jv
1	0.01450224648	0.019333638	52344	bank
2	0.03981890107	0.024530033	52344	bank
3	0.05228616775	0.028503744	52344	bank
4	0.05834133032	0.041151635	52343	bank
5	0.06178151902	0.036278538	52345	bank
6	0.06353438436	0.066336893	79247	bank
7	0.06456147817	0.11128582	25439	bank
8	0.06870304711	0.081284977	52359	bank
9	0.0742767586	0.084809572	52329	bank
10	0.1156135502	0.14242592	52343	bank

Subgroup Results

AUC improved across all reported age, gender, income and bank-exposure groups. The largest improvement occurred among borrowers with no recorded bank credit limit or balance, where AUC increased from 0.555 to 0.759.

This suggests that JV information is particularly useful when bank information is limited. These subgroup comparisons are exploratory and do not include confidence intervals for differences between groups.

subgroup_comparison

Uncertainty Around the Findings

The paired bootstrap used 1,000 repetitions, resampling the same test borrowers for both models in each repetition. All ten reported performance improvements had 95% confidence intervals that excluded zero, supporting the finding that adding JV information improves prediction.

The estimated AUC improvement was 0.0947, with a 95% confidence interval of 0.0922 to 0.0970. The updated results are consistent with the initial bootstrap findings.

These intervals reflect uncertainty from test-borrower sampling, conditional on the fitted models. They do not include uncertainty from refitting the models or establishing how performance would change in another period or borrower population.

metric	point_difference	ci_lower	ci_upper	bootstrap_sd	repetitions	improvement	improvement_ci_lower	improvement_ci_upper
auc	0.0947022203	0.09221909743	0.09695382827	0.001239757817	1000	0.0947022203	0.09221909743	0.09695382827
average_precision	0.05738822637	0.05540168638	0.05948102826	0.001057470674	1000	0.05738822637	0.05540168638	0.05948102826
log_loss	-0.01229967475	-0.01291832797	-0.01169856866	0.000300186216	1000	0.01229967475	0.01169856866	0.01291832797
brier	-0.001907290334	-0.001971815367	-0.001842668167	0.000032800460 52	1000	0.001907290334	0.001842668167	0.001971815367
recall_fpr05	0.07556580834	0.07162718922	0.08002045014	0.002088053018	1000	0.07556580834	0.07162718922	0.08002045014
recall_fpr10	0.09969449467	0.09455762054	0.10477107	0.002634197944	1000	0.09969449467	0.09455762054	0.10477107
precision_top05	0.06372128281	0.05986386581	0.06804327064	0.002118650159	1000	0.06372128281	0.05986386581	0.06804327064
lift_top05	1.039780445	0.9765430098	1.109749504	0.03433250512	1000	1.039780445	0.9765430098	1.109749504
precision_top10	0.05269090423	0.05018818 18	0.05515589477	0.001296083832	1000	0.05269090423	0.05018818 18	0.05515589477
lift_top10	0.8597907861	0.8188636182	0.8997809513	0.020718537 17	1000	0.8597907861	0.8188636182	0.8997809513

Conclusion

JV NBI credit information improves default prediction across several measures. The strongest improvement comes from including credit limit and balance together. The expanded model identifies more defaulters at comparable screening constraints, with particularly large gains among borrowers who have limited bank information.

The remaining priority is calibration: improved prediction does not mean that estimated default probabilities are accurate in every risk group. The findings demonstrate predictive value, rather than a causal effect on repayment or a measured increase in lending profitability.